Carlos Henrique Aranguren

cha4@stanford.edu | linkedin.com/in/carlos-henrique-aranguren | Stanford, CA

Education

Stanford University — Stanford, CA
M.S. in Electrical Engineering — *December 2027*

- **GPA**: 3.87/4.00
- **Coursework**: Digital Systems Architecture (EE 180), Intro to Mechatronics (EE 118), Computer Systems from the Ground Up (CS 107E), Linear Dynamical Systems (EE 263), Signals and Systems (EE 102A), Deep Learning for Computer Vision (CS 231N)

Experience

Kawasaki Heavy Industries — Akashi, Japan
Advanced System Technology Intern — *June 2026 – August 2026*

- Trained a teacher-student reinforcement-learning locomotion policy for rough-terrain quadruped walking in NVIDIA Isaac Lab, reaching a 95% episode success rate under heavy sensor occlusion and noise.
- Closed the sim-to-real gap by tuning action scaling, default joint positions, and LiDAR point extraction for the policy's terrain perception.
- Deployed the first perception-based policy on an open-source quadruped platform that had previously run only blind policies, and eliminated gait jitter during hardware trials.

Krill Energy — Caracas, Venezuela
Integration Intern — *July 2025 – August 2025*

- Integrated and commissioned 5 solar microgrid systems for 3 major telecommunications clients in Venezuela, totaling 168 kWh of storage (7 × 4.8 kWh lithium modules per system) with charge controllers and inverters.
- Configured RS-485 communication between batteries, inverters, and controllers, and set up battery management logic and system configurations.
- Diagnosed interoperability faults during end-to-end testing using oscilloscopes and multimeters.

Stanford Robotics Center — Stanford, CA
Research Assistant — *April 2024 – December 2024*

- Redesigned joint components of the OceanOne underwater robot's manipulator arm to simplify fabrication and make the arm easier to build and replicate.
- Designed a steel crossbar mount with two end-point supports in OnShape to secure a 20 kg battery to the side of a mobile base, alongside a PC and Intel NUC.

Projects

Quadcopter Flight Controller | *C, RISC-V, Bare-Metal Firmware, PID Control* — September 2024 – December 2024

- Wrote bare-metal flight control firmware in C for a RISC-V Mango Pi board, with real-time IMU polling and PID attitude control.
- Brought up I2C sensor and ESC interfaces, using an oscilloscope to verify signal timing.
- Designed and 3D-printed the frame, housing, and redesigned propellers in OnShape to fix insufficient lift, achieving stable hover.

Line-Following Robot (EE 118) | *C++, KiCad, OnShape, Prototyping* — January 2024 – March 2024

- Built a line-following robot from scratch: drew KiCad schematics for the IR sensor array and motor drivers, and fabricated the chassis with laser cutting and 3D printing.
- Implemented a PD controller in C++ for line tracking; placed 1st of 20 teams as the fastest to complete the required course.

Technical Skills

Languages: C, C++, Python, RISC-V Assembly, MATLAB, Julia, Verilog
Embedded & Hardware: Bare-metal firmware, feedback control (PID/PD), KiCad schematic capture and PCB design, LTSpice, oscilloscopes, soldering
Robotics & ML: NVIDIA Isaac Lab, reinforcement learning, ROS2, PyTorch, NumPy
Tools & Fabrication: Linux, Git, OnShape CAD, 3D printing, laser cutting